conversation in real time—all they're capable of is detecting when a person is speaking, and the length of that speech.

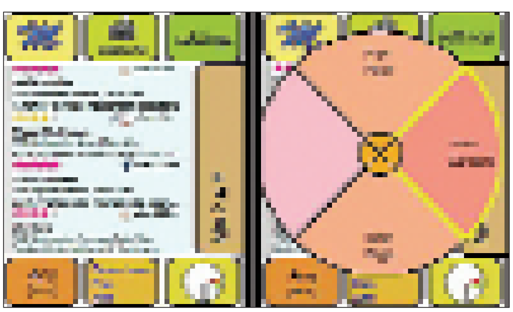
It's also unlikely that the software will know when to interrupt you and when not to—some initial setting up will be required, though it'll be a small price to pay for a system that'll make up for your lack of cellular etiquette, even though it'll take years to come.

And then, there's the matter that started us along these lines in the first place...

I'm Bored!

How'd you like a cell phone that told you what you could do with all that idle time? Or if you're walking around in a new city without a clue? PARC is working on a program called Magitti, that uses what it knows about you to recommend activities that you can enjoy in your vicinity.

What you'll see on your screen when you fire up Magitti is simple enough—a list of things to do in the area. When you first use it, this list won't be prioritised according to your likes, but as the software learns more about you, it'll be able to tell which recommendations are more likely to pique your interest—much like recommendation systems on sites like Amazon. For example, if you're a museum buff, historical landmarks will find themselves at the top of your list. It'll also be able to tailor its recommendations based on the time of the day—you can't go museum-ing late at night, so local pubs might be brought to the top



Magitti's user interface on a Windows Mobile-based phone

of the recommendation list. Magitti will use GPS to infer what choices you make, and find out more about your destination using an online database (which may eventually turn out to be the Wikipedia, for all we know).

In addition to learning from your choices, Magitti will also analyse the data on your cell phone—text messages, for instance—and like Google AdSense, make recommendations. If a friend messages you to meet for an inexpensive Chinese dinner, for example, Magitti will prioritise such places when you ask for recommendations. Extrapolating a bit, we anticipate Magitti-like software that will integrate with your Orkut, Facebook or other profile, and prioritise activities that your friends are recommending. The software will go through trials in Japan in the next few months, and we'll be watching eagerly.

Magitti may be able to learn from your choices, but what if it could know you by sensing your behaviour?

Beyond The Cell Phone

While the cell phone is the prime candidate for any experiment with context awareness, it isn't alone. At CeBIT 2007, we saw NEC's Dew concept camera that you can wear around your neck while it monitors your mood using the tone of your voice. If it detects happiness, it takes a picture, saving your happy moment forever (or till its memory runs out).

Closer to the shelves, we see the Sony CyberShot T200, which has a "Smile detection mode", which, well, detects smiles. Switch the camera to that mode, place it wherever you want, and it'll take a photo every time a smiling face is in its frame.

We anticipate Magitti-like software that will integrate with your Orkut, Facebook or other profile, and prioritise activities that your friends are recommending

Sensors And Sensibility

The iPhone, the Sony Ericsson W910i, the Nokia 5500 Sport and a host of soon-to-be-released phones sport accelerometers that enable various features depending on the device's movement. The iPhone and iPod Touch use it to rotate movies when you tilt the phone, the Sony Ericsson phones use shakes to shuffle tracks, and the 5500 uses it to measure the distance you've travelled and in a game where you tilt the phone to get a ball out of a maze

The iPhone also has an ambient light sensor that adjust the screen's brightness based on the brightness of its environment, and an infrared sensor that can tell when you're pressing the phone to your face. But why must we know this?

Nathan Eagle and Sandy Pentland of MIT's Media Laboratory lament that while new mobile phones are loaded with more sensors, they're rarely used for anything beyond trivial gimmicks. Collecting the data from these sensors could yield a whole lot of information about the user, and will enable software like Magitti work even more like magic. The data from an accelerometer in your phone, for example, can be used to tell whether you lead an active or sedentary lifestyle. It can also be used to tell whether you're walking, running, cycling or riding in a vehicle. Combined with GPS and Intel's "polite phone" software, your phone can collect data about where you go, what social situations you're in, who you talk to for how long, and so on. Eagle and Petland have even developed an algorithm that can predict your future actions based on the data your phone collects about you!

Give all that data (and the predictions) to software like Magitti, and you've got yourself a device that knows you so incredibly well that the mind boggles.

Brakes

Every great fantasy is thwarted by reality. In this case, it's the hardware—nearly all these concepts demand a lot more processing power than current phones can deliver, and even if we see such a powerhouse within the next year, processing all this data in real-time is bound to be murder on battery technology that's already showing wrinkles.

For now, we must content ourselves with GPS and (ugh) Wikimapia.

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